Figure S1. Kernel density plots of soil health ratings faceted by biological, chemical and physical classes. Ratings were centered based on *z*-score. Water capacity refers to 'available water capacity.'

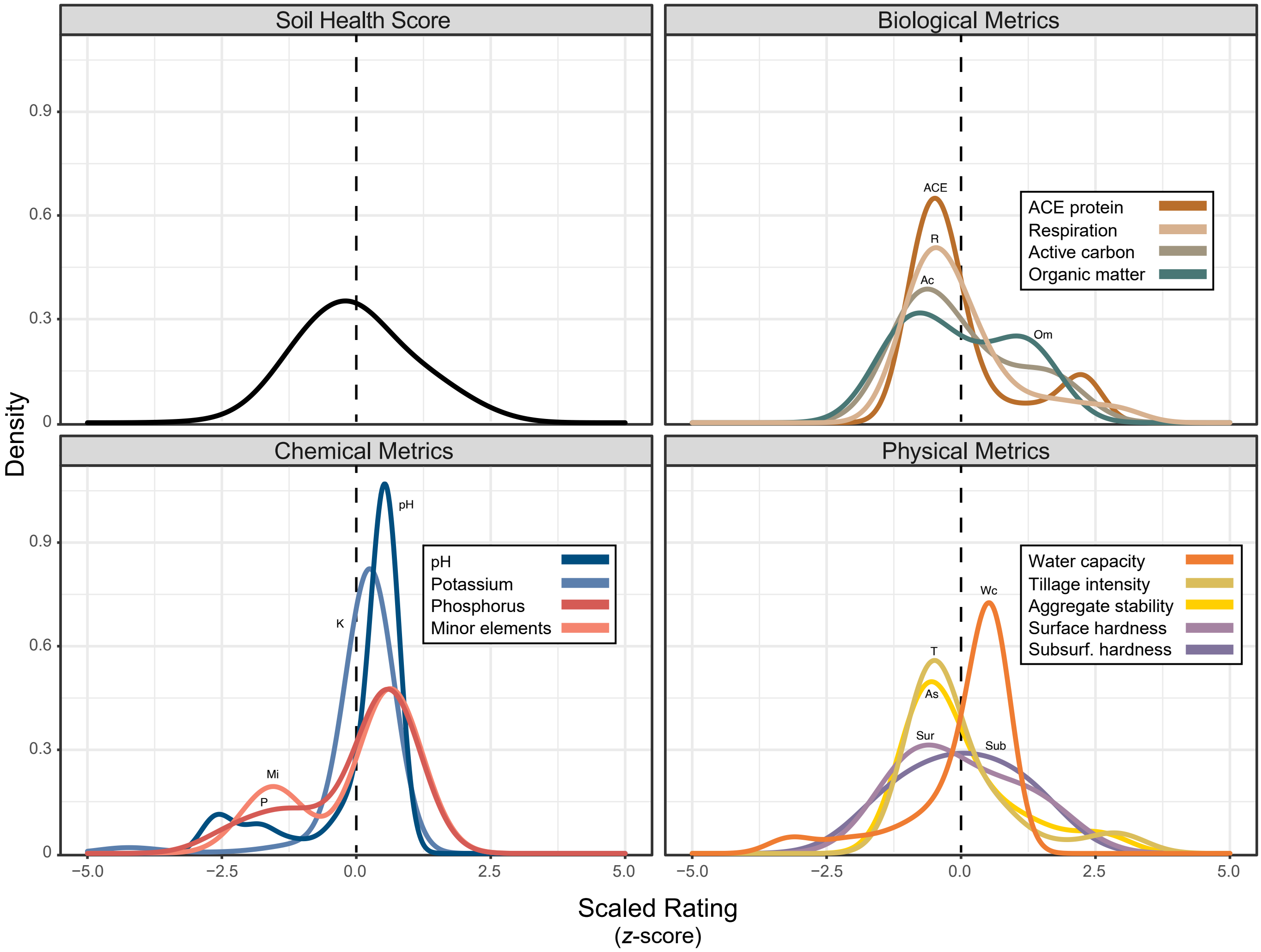

Figure S2. A complete ranking of the accuracy of ASV-based models for predicting health ratings (AB) and health categories (C), as well as soil texture, tillage and DNA yields. The results from SVM models are shown on the left and RF on the right. RF-based classification was not performed since it was outperformed by SVM and had long training times. Models for potassium and minor elements had low accuracy (not shown). Water capacity refers to 'available water capacity.'

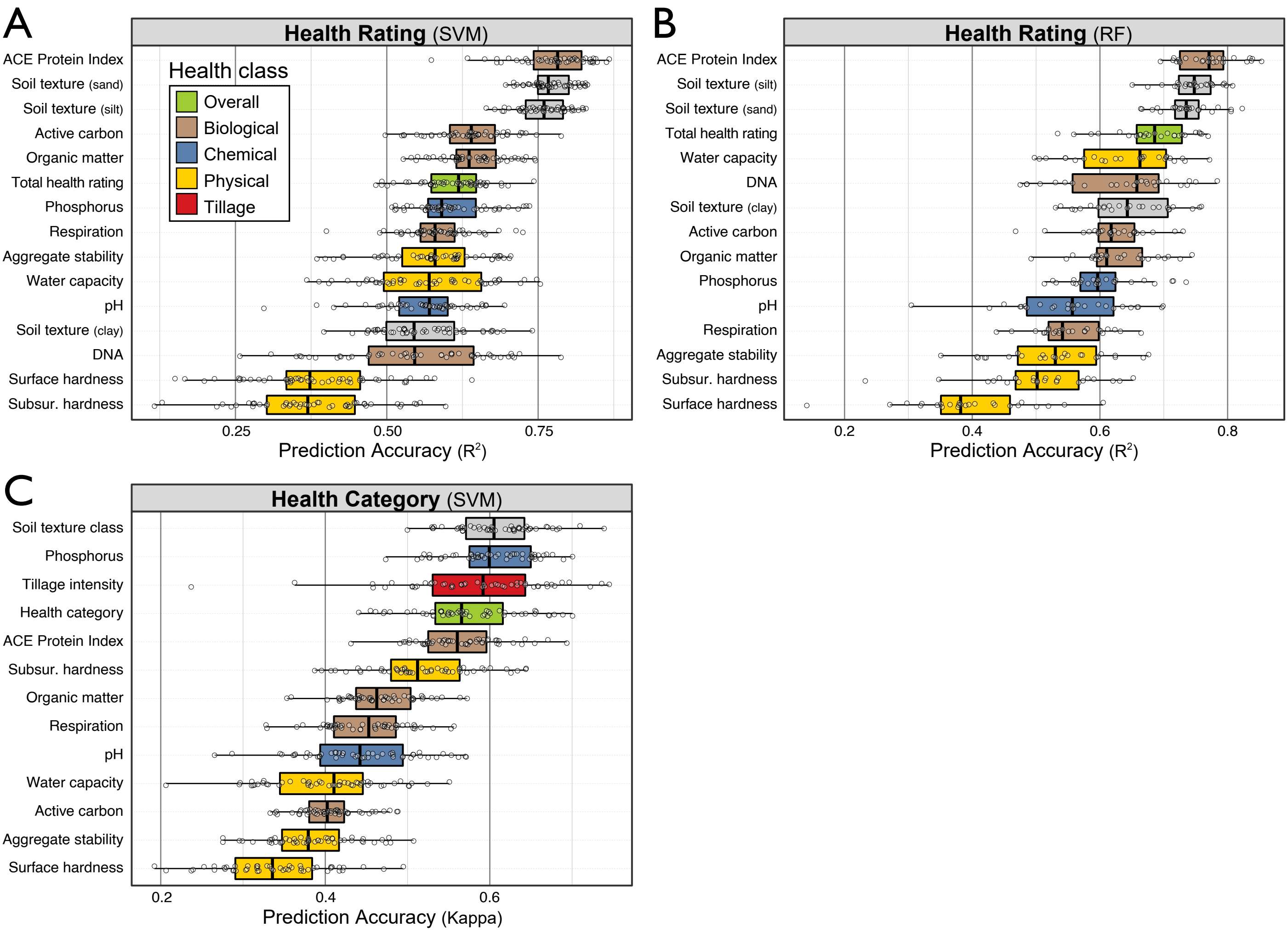

Figure S3. The accuracy (kappa) of classification-based models was better than when classifications were assigned post hoc to ratings predicted by regression-based models. In (A), histograms represent the frequency of kappa statistics derived from classification- (light brown) and regression-based (darker brown) SVM models. The superiority of classification-based models was apparent for all health metrics. In (B), the confusion matrices for classification- and regression-based predictions of health categories show the poorer accuracy of regression-based predictions for the extreme categories (highlighted in green and red). Each cell contains the average value for 50 models.